

Chapter 4

ARM Arithmetic and Logic Instructions

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Adding Two Integers

```
int x = 1;  
int y = 2;  
int z;
```

C Statement

```
z = x + y;
```

If values are in registers

- ▶ Value of **x** in **r0**
- ▶ Value of **y** in **r1**
- ▶ Value of **z** in **r2**

Assembly Statement

?

Adding Two Integers

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C Statement

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If values are in registers

- ▶ Value of **x** in **r0**
- ▶ Value of **y** in **r1**
- ▶ Value of **z** in **r2**

Assembly Statement

```
ADD r2, r1, r0
```

Destination

Source Operand 2

Source Operand 1

Adding Two Integers

```
uint x = 1;  
uint y = 2;  
uint z;
```

C Statement

```
z = x + y;
```

If values are in registers

- ▶ Value of **x** in **r0**
- ▶ Value of **y** in **r1**
- ▶ Value of **z** in **r2**

Assembly Statement

?

Adding Two Integers

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uint x = 1;  
uint y = 2;  
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C Statement

```
z = x + y;
```

If values are in registers

- ▶ Value of **x** in **r0**
- ▶ Value of **y** in **r1**
- ▶ Value of **z** in **r2**

Assembly Statement

```
ADD r2, r1, r0
```

ADD works for both signed and unsigned add operations.

Adding Two Integers

```
int x = 1;  
int y = 2;  
int z;
```

C Statement

```
z = x + y;
```

If addresses are in registers

- ▶ Address of x in r0
- ▶ Address of y in r1
- ▶ Address of z in r2

Assembly Statements

?

Adding Two Integers

```
int x = 1;  
int y = 2;  
int z;
```

C Statement

```
z = x + y;
```

If addresses are in registers

- ▶ Address of x in r0
- ▶ Address of y in r1
- ▶ Address of z in r2

Assembly Statements

```
LDR r3, [r0] ; Read x  
LDR r4, [r1] ; Read y  
ADD r5, r3, r4  
STR r5, [r2] ; Write z
```

Load, modify, and store

Example Arithmetic Instructions

- ▶ **ADD** `r0, r1, r2` ; $r0 = r1 + r2$
- ▶ **ADC** `r0, r1, r2` ; Add with carry, $r0 = r1 + r2 + \text{carry}$
- ▶ **SUB** `r0, r1, r2` ; $r0 = r1 - r2$
- ▶ **SBC** `r0, r1, r2` ; Subtract with borrow, $r0 = r1 - r2 - (1 - \text{carry})$
- ▶ **MUL** `r0, r1, r2` ; $r0 = r1 * r2$, product limited to 32 bits
- ▶ **UDIV** `r0, r1, r2` ; Unsigned divide, $r0 = r1 / r2$
- ▶ **SDIV** `r0, r1, r2` ; Signed divide, $r0 = r1 / r2$
- ▶ **SMULL** `r0, r1, r2, r3` ; Signed multiply (64-bit product), $r1:r0 = r2 * r3$
- ▶ **UMULL** `r0, r1, r2, r3` ; Unsigned multiply (64-bit product), $r1:r0 = r2 * r3$

Example Logical Instructions

- ▶ **AND** r0, r1, r2 ; Bitwise AND, $r0 = r1 \text{ AND } r2$
- ▶ **ORR** r0, r1, r2 ; Bitwise OR, $r0 = r1 \text{ OR } r2$
- ▶ **EOR** r0, r1, r2 ; Bitwise Exclusive OR, $r0 = r1 \text{ EOR } r2$
- ▶ **ORN** r0, r1, r2 ; Bitwise OR NOT, $r0 = r1 \text{ ORN } r2$
- ▶ **BIC** r0, r1, r2 ; Bit clear, $r0 = r1 \& \sim r2$

AND r0, r1, r2

32 bits

r1	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
r2	1	0	1	0	1	0	1	0	1	0	1	0	1	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	1
r0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit-wise Logic **AND**

Example Shift & Rotate Instructions

- ▶ **LSL** `r0, r1, r2` ; Logical shift left,
 $r0 = r1 \ll r2$
- ▶ **LSR** `r0, r1, r2` ; Logical shift right,
 $r0 = r1 \gg r2$
- ▶ **ASR** `r0, r1, r2` ; Arithmetic shift right,
 $r0 = r1 \ggg r2$
- ▶ **ROR** `r0, r1, r2` ; Rotate right,
 $r0 = r1 \text{ rotate by } r2 \text{ bits}$
- ▶ **RRX** `r0, r1, r2` ; Extended rotate right,
 $\{C, r0\} = \{C, r1\} \text{ rotate by } r2 \text{ bits}$

Logical Shift Left (**LSL**)



Logical Shift Right (**LSR**)



Arithmetic Shift Right (**ASR**)



Rotate Right (**ROR**)



Rotate Right Extended (**RRX**)



Example Data Transfer Instructions

- ▶ **MOV** r0, r1 ; Move, r0 = r1
- ▶ **MVN** r0, r1 ; Move NOT, r0 = bitwise NOT r1

MVN r0, r1

r1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
r0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0

Bit-wise Logic **NOT**

Overview:

Arithmetic and Logic Instructions

- ▶ **Shift** : **LSL** (logic shift left), **LSR** (logic shift right), **ASR** (arithmetic shift right), **ROR** (rotate right), **RRX** (rotate right with extend)
- ▶ **Logic**: **AND** (bitwise and), **ORR** (bitwise or), **EOR** (bitwise exclusive or), **ORN** (bitwise or not), **MVN** (move not)
- ▶ **Bit set/clear**: **BFC** (bit field clear), **BFI** (bit field insert), **BIC** (bit clear), **CLZ** (count leading zeroes)
- ▶ **Bit/byte reordering**: **RBIT** (reverse bit order in a word), **REV** (reverse byte order in a word), **REV16** (reverse byte order in each half-word independently), **REVSH** (reverse byte order in each half-word independently)
- ▶ **Addition**: **ADD**, **ADC** (add with carry)
- ▶ **Subtraction**: **SUB**, **RSB** (reverse subtract), **SBC** (subtract with carry)
- ▶ **Multiplication**: **MUL** (multiply), **MLA** (multiply-accumulate), **MLS** (multiply-subtract), **SMULL** (signed long multiply-accumulate), **SMLAL** (signed long multiply-accumulate), **UMULL** (unsigned long multiply-subtract), **UMLAL** (unsigned long multiply-subtract)
- ▶ **Division**: **SDIV** (signed), **UDIV** (unsigned)
- ▶ **Saturation**: **SSAT** (signed), **USAT** (unsigned)
- ▶ **Sign extension**: **SXTB** (signed), **SXTH**, **UXTB**, **UXTH**
- ▶ **Bit field extract**: **SBFX** (signed), **UBFX** (unsigned)
- ▶ Syntax

<Operation>{<cond>}{S} Rd, Rn, Operand2

Example: Add

- ▶ Unified Assembler Language (UAL) Syntax

ADD r1, r2, r3 ; r1 = r2 + r3

ADD r1, r2, #4 ; r1 = r2 + 4

- ▶ Traditional Thumb Syntax

ADD r1, r3 ; r1 = r1 + r3

ADD r1, #15 ; r1 = r1 + 15

Commonly Used Arithmetic Operations

ADD {Rd,} Rn, Op2	Add $Rd \leftarrow Rn + Op2$
ADC {Rd,} Rn, Op2	Add with carry $Rd \leftarrow Rn + Op2 + Carry$
SUB {Rd,} Rn, Op2	Subtract $Rd \leftarrow Rn - Op2$
SBC {Rd,} Rn, Op2	Subtract with carry $Rd \leftarrow Rn - Op2 + Carry - 1$
RSB {Rd,} Rn, Op2	Reverse subtract $Rd \leftarrow Op2 - Rn$
MUL {Rd,} Rn, Rm	Multiply $Rd \leftarrow (Rn \times Rm)[31:0]$
MLA Rd, Rn, Rm, Ra	Multiply with accumulate $Rd \leftarrow (Ra + (Rn \times Rm))[31:0]$
MLS Rd, Rn, Rm, Ra	Multiply and subtract $Rd \leftarrow (Ra - (Rn \times Rm))[31:0]$
SDIV {Rd,} Rn, Rm	Signed divide $Rd \leftarrow Rn \div Rm$
UDIV {Rd,} Rn, Rm	Unsigned divide $Rd \leftarrow Rn \div Rm$
SSAT Rd, #n, Rm {,shift #s}	Signed saturate
USAT Rd, #n, Rm {,shift #s}	Unsigned saturate



ARM Programming Model

R0
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13: Stack Pointer (SP)
R14: Link Register (LR)
R15: Program Counter (PC)

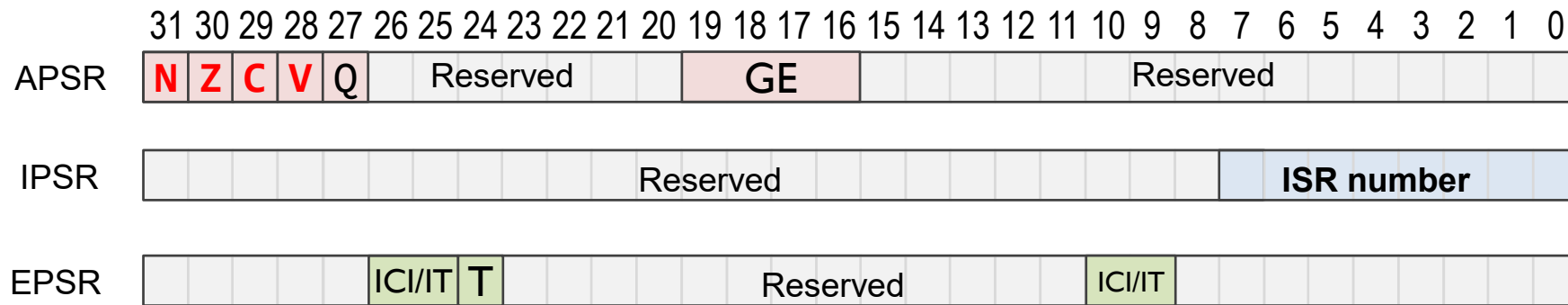


CPSR (Current Program Status Register)

- Four flag bits:
 - N (negative), Z (zero), C (carry), V (overflow).

Program Status Register (PSR)

- ▶ Application PSR (**APSR**), Interrupt PSR (**IPSR**), Execution PSR (**EPSR**)



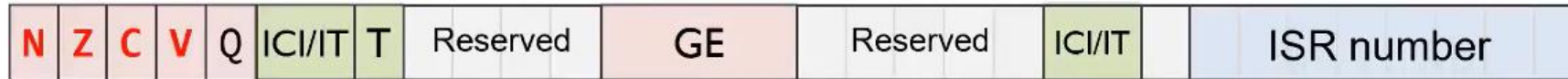
Combine them together into one register (**PSR**)



Note:

- GE flags are only available on Cortex-M4 and M7
- Use PSR in code

NZCV Flags in xPSR



- ▶ **N**: I/O = Result from ALU is **N**egative/positive
- ▶ **Z**: I/O = Result from ALU is **Z**ero/non-zero
- ▶ **C**: Three cases:
 - ▶ I/O = ALU addition **C**arry out/no carry out
 - ▶ I/O = ALU subtraction no borrow/borrow
 - ▶ I/O = Bit shifted/rotated out
- ▶ **V**: I/O = ALU o**V**erflowed/no overflow

Borrow and carry share the same flag bit.
For unsigned subtract,
Borrow = NOT Carry

Updating NZCV flags in PSR

Flags not changed		Flags updated
ADD	→	ADD S
SUB	→	SUB S
MUL	→	MUL S
UDIV	→	UDIV S
AND	→	AND S
ORR	→	ORR S
LSL	→	LSL S
MOV	→	MOV S

`CMP r1, r2` vs `SUBS r0, r1, r2`

Some instructions update NZCV flags even if no S is specified.

- **CMP**: Compare, like SUBS but without destination register
- **CMN**: Compare Negative, like ADDS but without destination register
- **TST**: Test, like ANDS but without destination register
- **TEQ**: Test equivalence, like EORS but without destination register

*Most instructions update NZCV flags
only if S suffix is present*

ADD *vs* ADDS

ADD r0, r1, r2 ; r0 = r1 + r2, NZCV flags unchanged

ADDS r0, r1, r2 ; r0 = r1 + r2, NZCV flags updated

- ▶ ADD does not update flags
- ▶ ADDS updates flags
 - ▶ xPSR.**N** = bit 31 of result
 - ▶ xPSR.**Z** = IsZero(result)
 - ▶ xPSR.**C** = carry, assuming r1 and r2 representing unsigned integers
 - ▶ xPSR.**V** = overflow, assuming r1 and r2 representing signed integers

Suffix S: Update Flags

```
LDR  r0, =0xFFFFFFFF
LDR  r1, =0x00000001
ADDS r0, r0, r1
```

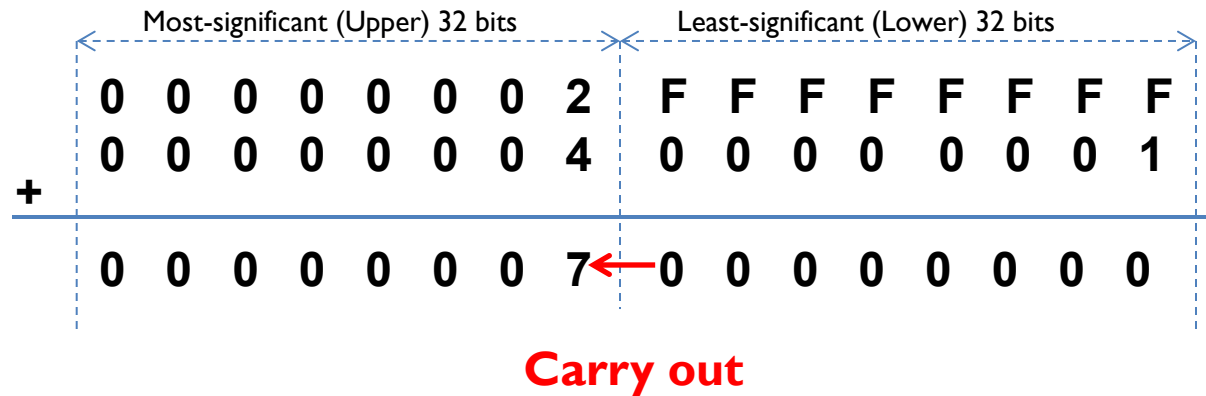
0xFFFFFFFF	r0
+ 0x00000001	r1
0x00000000	sum

N (Negative) = 0
Z (Zero) = 1
C (Carry) = 1
V (oVerflow) = 0

The screenshot displays the Keil uVision IDE interface. On the left, the 'Registers' window shows the state of the Cortex-M3 registers. The 'Core' registers R0 through R15 are visible, with R0 containing 0xFFFFFFFF and R15 (PC) containing 0x0800136. The 'xPSR' register shows the status of the flags: N=0, Z=1, C=1, V=0. On the right, the 'Disassembly' window shows the assembly code for the 'main.s' file. The code includes a comment block, an include statement for 'stm32l1xx_constants.s', and the definition of the 'main' function. The assembly code for the 'main' function is as follows:

```
1 ;***** (C) Yifeng ZHU *****
2 ; @file    main.s
3 ; @author  Yifeng Zhu
4 ;*****
5
6         INCLUDE stm32l1xx_constants.s
7
8         AREA    main, CODE, READONLY
9         EXPORT  __main
10        ENTRY
11
12        __main
13        PROC
14
15        LDR  r0, =0xFFFFFFFF
16        LDR  r1, =0x00000001
17        ADDS r3, r0, r1
18
19        stop
20
21        ENDP
22        ALIGN
23        END
```

Example: 64-bit Addition



- A register can only store 32 bits
- A 64-bit integer needs two registers
- Split 64-bit addition into two 32-bit additions

Example: 64-bit Addition

start

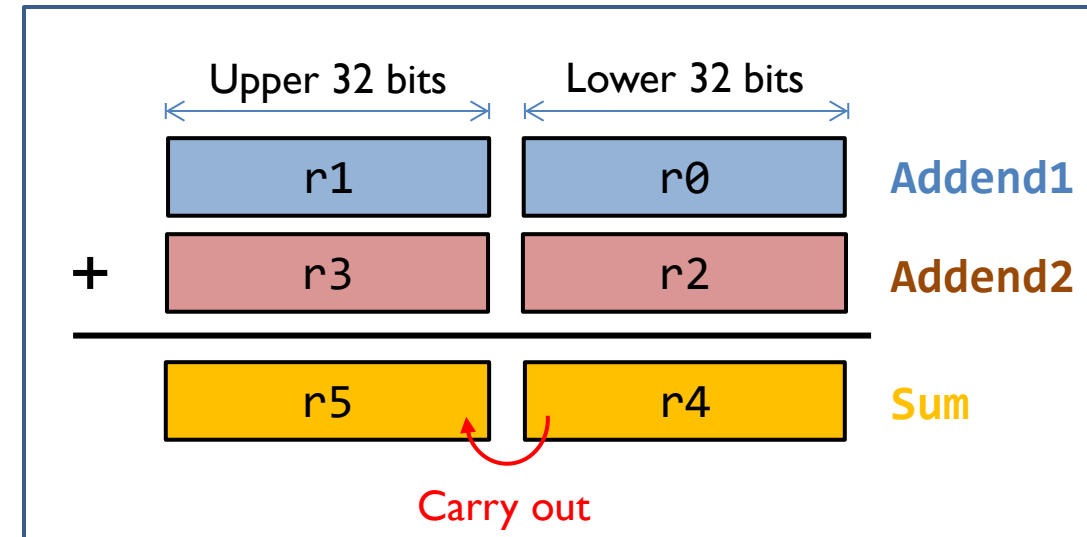
```
; C = A + B
; Two 64-bit integers A (r1,r0) and B (r3,r2).
; Result C (r5, r4)
; A = 00000002FFFFFFFF
; B = 0000000400000001
LDR  r0, =0xFFFFFFFF ; A's lower 32 bits
LDR  r1, =0x00000002   ; A's upper 32 bits
LDR  r2, =0x00000001   ; B's lower 32 bits
LDR  r3, =0x00000004   ; B's upper 32 bits
```

```
; Add A to B
```

```
ADDS r4, r2, r0 ; C[31..0] = A[31..0] + B[31..0], update Carry
```

```
ADC  r5, r3, r1 ; C[64..32] = A[64..32] + B[64..32] + Carry
```

stop B stop



Example: 64-bit Subtraction

start

```
; C = A - B
; Two 64-bit integers A (r1,r0) and B (r3,r2).
; Result C (r5, r4)
; A = 00000002FFFFFFFF
; B = 0000000400000001
```

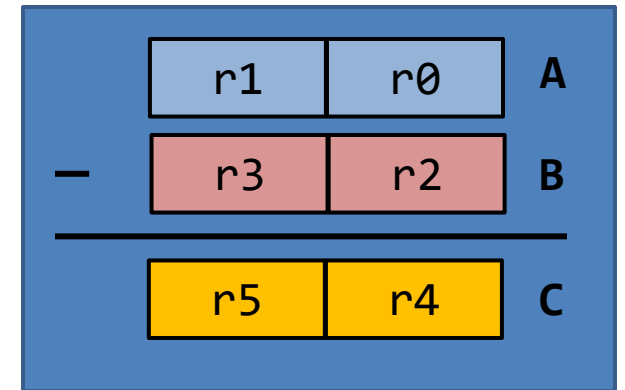
```
LDR  r0, =0xFFFFFFFF ; A's lower 32 bits
LDR  r1, =0x00000002  ; A's upper 32 bits
LDR  r2, =0x00000001  ; B's lower 32 bits
LDR  r3, =0x00000004  ; B's upper 32 bits
```

```
; Subtract B from A
```

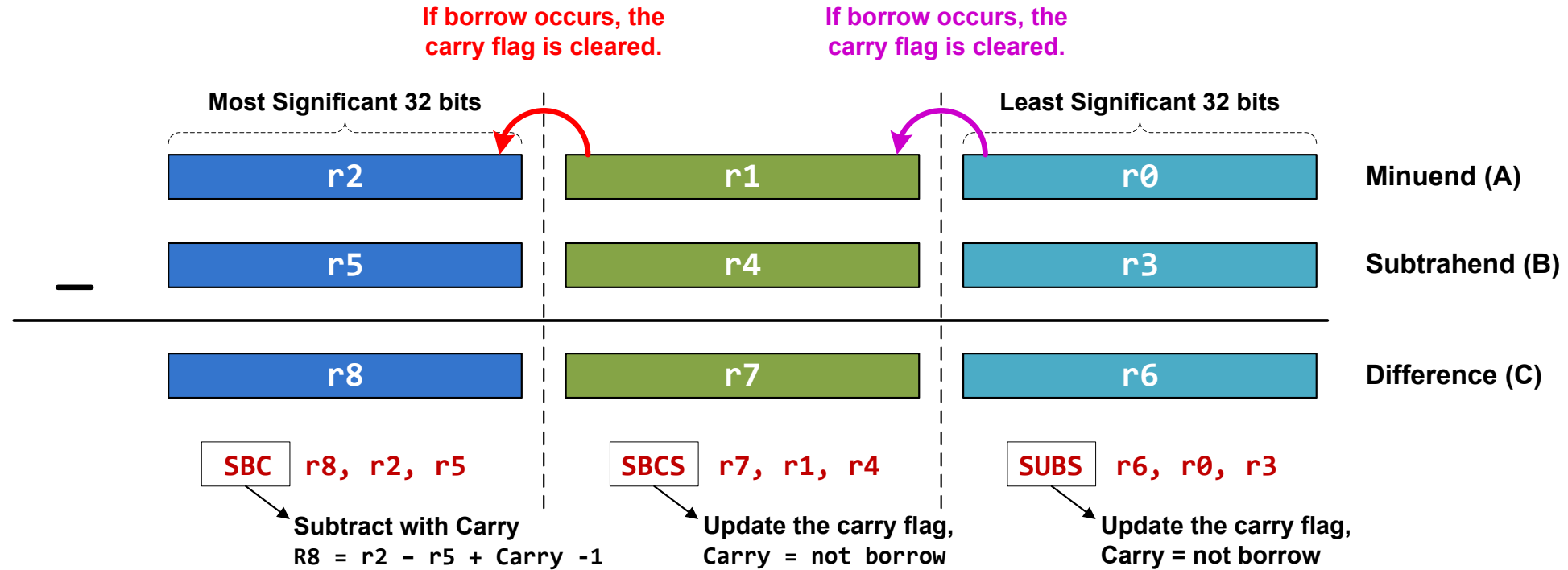
```
SUBS r4, r0, r2 ; C[31..0] = A[31..0] - B[31..0], update Carry
```

```
SBC  r5, r1, r3 ; C[64..32] = A[64..32] - B[64..32] - (1 - Carry)
```

stop B stop



Example: 96-bit Subtraction



SUBS r6, r0, r3

SBCS r7, r1, r4

SBC r8, r2, r5

Example: Short Multiplication and Division

MUL: Signed multiply

MUL r6, r4, r2 ; r6 = LSB32(r4 × r2)

UMUL: Unsigned multiply

UMUL r6, r4, r2 ; r6 = LSB32(r4 × r2)

MLA: Multiply with accumulation

MLA r6, r4, r1, r0 ; r6 = LSB32(r4 × r1) + r0

MLS: Multiply with subtract

MLS r6, r4, r1, r0 ; r6 = LSB32(r4 × r1) - r0

LSB32: Least significant 32 bits

Example: Long Multiplication

UMULL RdLo, RdHi, Rn, Rm	Unsigned long multiply $\text{RdHi}, \text{RdLo} \leftarrow \text{unsigned}(\text{Rn} \times \text{Rm})$
SMULL RdLo, RdHi, Rn, Rm	Signed long multiply $\text{RdHi}, \text{RdLo} \leftarrow \text{signed}(\text{Rn} \times \text{Rm})$
UMLAL RdLo, RdHi, Rn, Rm	Unsigned multiply with accumulate $\text{RdHi}, \text{RdLo} \leftarrow \text{unsigned}(\text{RdHi}, \text{RdLo} + \text{Rn} \times \text{Rm})$
SMLAL RdLo, RdHi, Rn, Rm	Signed multiply with accumulate $\text{RdHi}, \text{RdLo} \leftarrow \text{signed}(\text{RdHi}, \text{RdLo} + \text{Rn} \times \text{Rm})$

The result has 64 bits, placed in two registers.

```
UMULL r3, r4, r0, r1    ; r4:r3 = r0 × r1, r4 = MSB bits, r3 = LSB bits
SMULL r3, r4, r0, r1    ; r4:r3 = r0 × r1
UMLAL r3, r4, r0, r1    ; r4:r3 = r4:r3 + r0 × r1
SMLAL r3, r4, r0, r1    ; r4:r3 = r4:r3 + r0 × r1
```

Bitwise Logic

AND {Rd,} Rn, Op2	Bitwise logic AND $Rd \leftarrow Rn \& \text{operand2}$
ORR {Rd,} Rn, Op2	Bitwise logic OR $Rd \leftarrow Rn \mid \text{operand2}$
EOR {Rd,} Rn, Op2	Bitwise logic exclusive OR $Rd \leftarrow Rn \wedge \text{operand2}$
ORN {Rd,} Rn, Op2	Bitwise logic NOT OR $Rd \leftarrow Rn \mid (\text{NOT } \text{operand2})$
BIC {Rd,} Rn, Op2	Bit clear $Rd \leftarrow Rn \& \text{NOT } \text{operand2}$
BFC Rd, #lsb, #width	Bit field clear $Rd[(\text{width}+\text{lsb}-1):\text{lsb}] \leftarrow 0$
BFI Rd, Rn, #lsb, #width	Bit field insert $Rd[(\text{width}+\text{lsb}-1):\text{lsb}] \leftarrow Rn[(\text{width}-1):0]$
MVN Rd, Op2	Move NOT, logically negate all bits $Rd \leftarrow 0xFFFFFFFF \text{ EOR } \text{Op2}$

Example: AND r2, r0, r1

32 bits

r0	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
r1	1	0	1	0	1	0	1	0	1	0	1	0	1	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	1
r2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit-wise Logic AND

Example: ORR r2, r0, r1

32 bits

r0	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
r1	1	0	1	0	1	0	1	0	1	0	1	0	1	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	1
r2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	

Bit-wise Logic OR

Example: BIC r2, r0, r1

Bit Clear

$$r2 = r0 \& \text{NOT } r1$$

Step 1:

r1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
NOT r1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0

Step 2:

r0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		
NOT r1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0		
r2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	

Example: **BFC** and **BFI**

- ▶ Bit Field Clear (**BFC**) and Bit Field Insert (**BFI**).

- ▶ Syntax

 - ▶ **BFC** Rd, #lsb, #width

 - ▶ **BFI** Rd, Rn, #lsb, #width

- ▶ Examples:

BFC R4, #8, #12

; Clear bit 8 to bit 19 (a total of 12 bits) of R4

BFI R9, R2, #8, #12

; Replace bit 8 to bit 19 (12 bits) of R9

; with bit 0 to bit 11 from R2.

Bit Operators (&, |, ~) vs Boolean Operators (&&, ||, !)

A && B	Boolean and	A & B	Bitwise and
A B	Boolean or	A B	Bitwise or
!B	Boolean not	~B	Bitwise not

- ▶ The Boolean operators perform word-wide operations, not bitwise.
- ▶ For example,
 - ▶ “0x10 & 0x01” = 0x00, but “0x10 && 0x01” = 0x01.
 - ▶ “~0x01” = 0xFFFFFFFFFE, but “!0x01” = 0x00.

Saturating Instruction: **SSAT** and **USAT**

- ▶ Syntax:

- ▶ `op{cond} Rd, #n, Rm{, shift}`

- ▶ **SSAT** saturates a signed value to the signed range $-2^{n-1} \leq x \leq 2^{n-1} - 1$.

$$SAT(x) = \begin{cases} 2^{n-1} - 1 & \text{if } x > 2^{n-1} - 1 \\ -2^{n-1} & \text{if } x < -2^{n-1} \\ x & \text{otherwise} \end{cases}$$

- ▶ **USAT** saturates a signed value to the unsigned range $0 \leq x \leq 2^n - 1$.

$$USAT(x) = \begin{cases} 2^n - 1 & \text{if } x > 2^n - 1 \\ x & \text{otherwise} \end{cases}$$

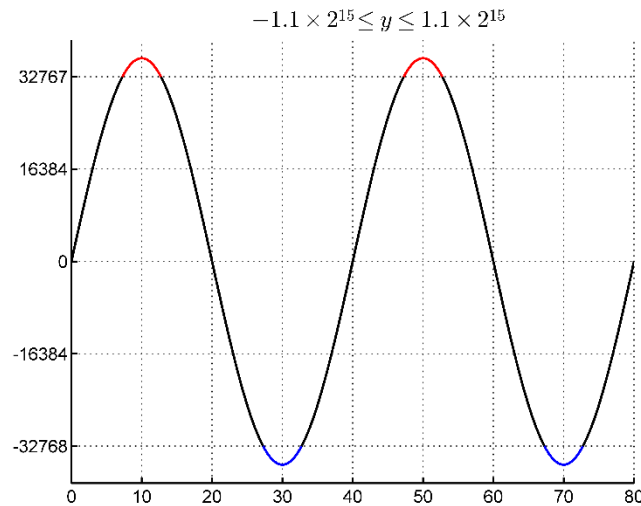
- ▶ Examples:

- ▶ `SSAT r2, #11, r1` ; output range: $-2^{10} \leq r2 \leq 2^{10}$

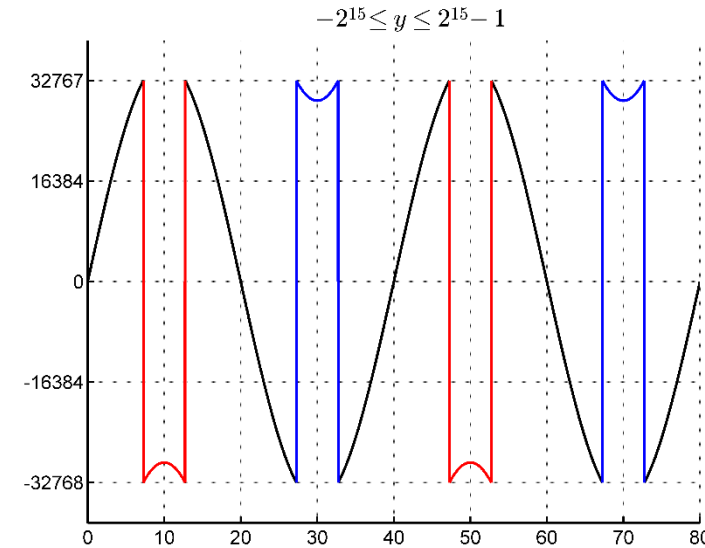
- ▶ `USAT r2, #11, r3` ; output range: $0 \leq r2 \leq 2^{11}$

Example of Saturation

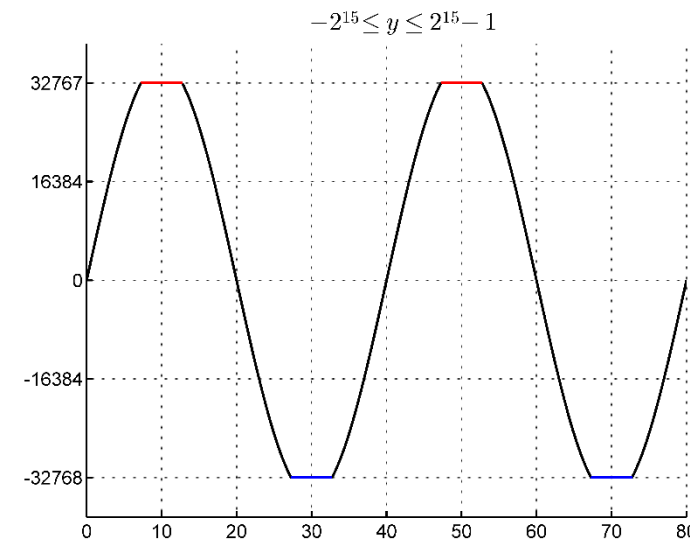
Assume data are limited to **16** bits



Without
saturation



With
saturation



Reverse Order

RBIT Rd, Rn	Reverse bit order in a word for (i = 0; i < 32; i++) Rd[i] ← RN[31- i]
REV Rd, Rn	Reverse byte order in a word Rd[31:24] ← Rn[7:0], Rd[23:16] ← Rn[15:8], Rd[15:8] ← Rn[23:16], Rd[7:0] ← Rn[31:24]
REV16 Rd, Rn	Reverse byte order in each half-word Rd[15:8] ← Rn[7:0], Rd[7:0] ← Rn[15:8], Rd[31:24] ← Rn[23:16], Rd[23:16] ← Rn[31:24]
REVSH Rd, Rn	Reverse byte order in bottom half-word and sign extend Rd[15:8] ← Rn[7:0], Rd[7:0] ← Rn[15:8], Rd[31:16] ← Rn[7] & 0xFFFF

RBIT Rd, Rn

Rn

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---

Rd

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
---	---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

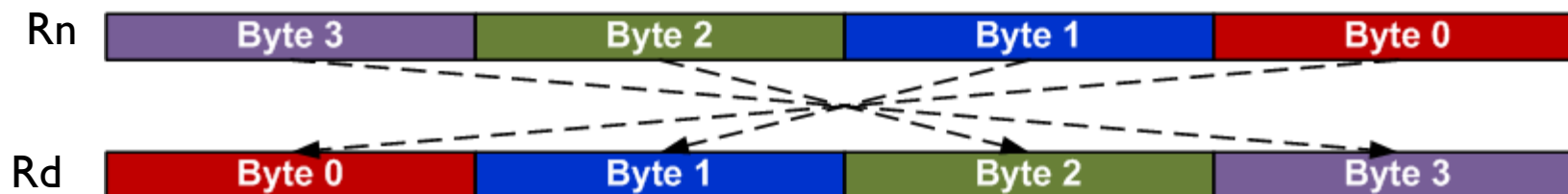
Example:

```
LDR  r0, =0x12345678 ; r0 = 0x12345678
RBIT r1, r0           ; Reverse bits, r1 = 0x1E6A2C48
```

Reverse Order

RBIT Rd, Rn	Reverse bit order in a word for (i = 0; i < 32; i++) Rd[i] ← RN[31- i]
REV Rd, Rn	Reverse byte order in a word Rd[31:24] ← Rn[7:0], Rd[23:16] ← Rn[15:8], Rd[15:8] ← Rn[23:16], Rd[7:0] ← Rn[31:24]
REV16 Rd, Rn	Reverse byte order in each half-word Rd[15:8] ← Rn[7:0], Rd[7:0] ← Rn[15:8], Rd[31:24] ← Rn[23:16], Rd[23:16] ← Rn[31:24]
REVSH Rd, Rn	Reverse byte order in bottom half-word and sign extend Rd[15:8] ← Rn[7:0], Rd[7:0] ← Rn[15:8], Rd[31:16] ← Rn[7] & 0xFFFF

REV Rd, Rn



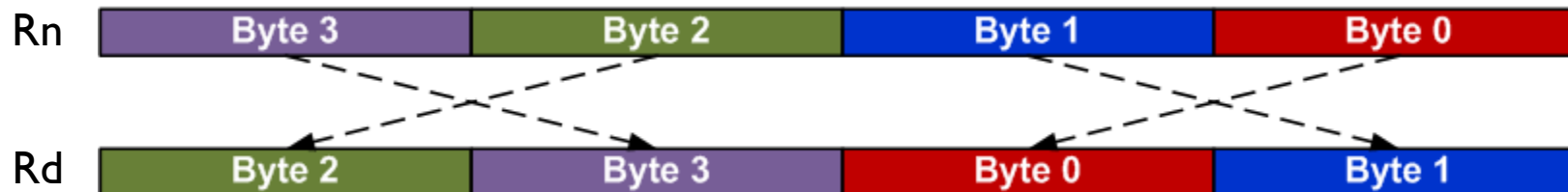
Example:

```
LDR R0, =0x12345678 ; R0 = 0x12345678
REV R1, R0           ; R1 = 0x78563412
```

Reverse Order

RBIT Rd, Rn	Reverse bit order in a word for (i = 0; i < 32; i++) Rd[i] ← RN[31- i]
REV Rd, Rn	Reverse byte order in a word Rd[31:24] ← Rn[7:0], Rd[23:16] ← Rn[15:8], Rd[15:8] ← Rn[23:16], Rd[7:0] ← Rn[31:24]
REV16 Rd, Rn	Reverse byte order in each half-word Rd[15:8] ← Rn[7:0], Rd[7:0] ← Rn[15:8], Rd[31:24] ← Rn[23:16], Rd[23:16] ← Rn[31:24]
REVSH Rd, Rn	Reverse byte order in bottom half-word and sign extend Rd[15:8] ← Rn[7:0], Rd[7:0] ← Rn[15:8], Rd[31:16] ← Rn[7] & 0xFFFF

REV16 Rd, Rn



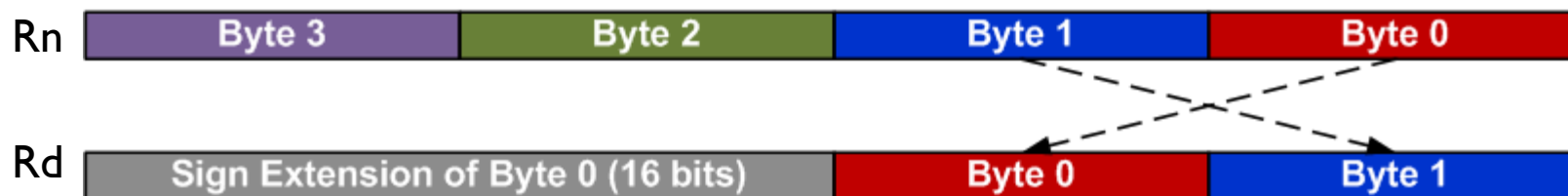
Example:

```
LDR R0, =0x12345678    ; R0 = 0x12345678
REV16 R2, R0             ; R2 = 0x34127856
```

Reverse Order

RBIT Rd, Rn	Reverse bit order in a word for (i = 0; i < 32; i++) Rd[i] ← RN[31- i]
REV Rd, Rn	Reverse byte order in a word Rd[31:24] ← Rn[7:0], Rd[23:16] ← Rn[15:8], Rd[15:8] ← Rn[23:16], Rd[7:0] ← Rn[31:24]
REV16 Rd, Rn	Reverse byte order in each half-word Rd[15:8] ← Rn[7:0], Rd[7:0] ← Rn[15:8], Rd[31:24] ← Rn[23:16], Rd[23:16] ← Rn[31:24]
REVSH Rd, Rn	Reverse byte order in bottom half-word and sign extend Rd[15:8] ← Rn[7:0], Rd[7:0] ← Rn[15:8], Rd[31:16] ← Rn[7] & 0xFFFF

REVSH Rd, Rn



Example:

```
LDR R0, =0x33448899    ; R0 = 0x33448899
REVSH R1, R0            ; R0 = 0xFFFF9988
```

Sign and Zero Extension

```
int8_t  a = -1;    // a signed 8-bit integer,  a = 0xFF
int16_t b = -2;    // a signed 16-bit integer, b = 0xFFFE
int32_t c;         // a signed 32-bit integer

c = a;             // sign extension required, c = 0xFFFFFFFF
c = b;             // sign extension required, c = 0xFFFFFFFFE
```

Sign and Zero Extension

SXTB {Rd,} Rm {,ROR #n}	Sign extend a byte $Rd[31:0] \leftarrow \text{Sign Extend}((Rm \text{ ROR } (8 \times n))[7:0])$
SXTH {Rd,} Rm {,ROR #n}	Sign extend a half-word $Rd[31:0] \leftarrow \text{Sign Extend}((Rm \text{ ROR } (8 \times n))[15:0])$
UXTB {Rd,} Rm {,ROR #n}	Zero extend a byte $Rd[31:0] \leftarrow \text{Zero Extend}((Rm \text{ ROR } (8 \times n))[7:0])$
UXTH {Rd,} Rm {,ROR #n}	Zero extend a half-word $Rd[31:0] \leftarrow \text{Zero Extend}((Rm \text{ ROR } (8 \times n))[15:0])$

LDR R0, =0x55AA8765

SXTB R1, R0 ; R1 = 0x00000065

SXTH R1, R0 ; R1 = 0xFFFF8765

UXTB R1, R0 ; R1 = 0x00000065

UXTH R1, R0 ; R1 = 0x00008765

Move Data between Registers

MOV	$Rd \leftarrow \text{operand2}$
MVN	$Rd \leftarrow \text{NOT operand2}$
MRS $Rd, \text{spec_reg}$	Move from special register to general register
MSR $\text{spec_reg}, Rm$	Move from general register to special register

MOV $r4, r5$	; Copy $r5$ to $r4$
MVN $r4, r5$	; $r4 = \text{bitwise logical NOT of } r5$
MOV $r1, r2, \text{LSL } \#3$	; $r1 = r2 \ll 3$
MOV $r0, PC$	; Copy PC ($r15$) to $r0$
MOV $r1, SP$	; Copy SP ($r14$) to $r1$

Move Immediate Number to Register

MOVW Rd, #imm16	Move Wide, $Rd \leftarrow \#imm16$
MOVT Rd, #imm16	Move Top, $Rd \leftarrow \#imm16 \ll 16$
MOV Rd, #const	Move, $Rd \leftarrow \text{const}$

Example: Load a 32-bit number into a register

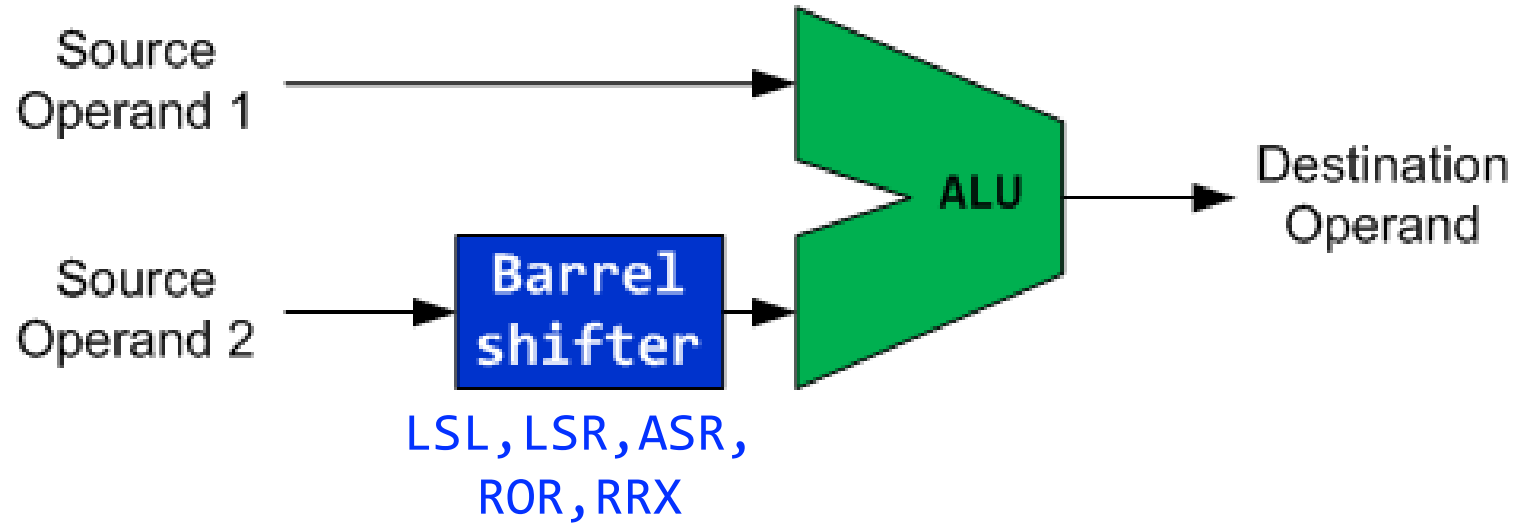
```
MOVW r0, #0x4321    ; r0 = 0x00004321
MOVT r0, #0x8765     ; r0 = 0x87654321
```

Order does matter!

- **MOVW** will zero the upper halfword
- **MOVT** won't zero the lower halfword

```
MOVT r0, #0x8765     ; r0 = 0x8765xxxx
MOVW r0, #0x4321     ; r0 = 0x00004321
```

Flexible 2nd Source Operand



ADD r0, r1, Operand2

- ▶ Add r0, r1, **r2** ; $r0 = r1 + r2$
- ▶ Add r0, r1, **#1** ; $r0 = r1 + 1$
- ▶ Add r0, r1, **r2 LSL #2** ; $r0 = r1 + r2 \ll 2$

Use Shifts To Implement Multiplication And Division

- ▶ Use Barrel shifter to speed up multiplication and division

- ▶ Shifting left 1 bit $\Leftrightarrow$ multiplication by 2

- ▶ Examples:

- ▶ $r1 = 9 \times r0 = r0 + 8 \times r0$

`ADD r1, r0, r0, LSL #3` $\Leftrightarrow$ `MOV r2, #9` ; $r2 = 9$

`MUL r1, r0, r2` ; $r1 = r0 * 9$

MUL instruction takes only registers, not an immediate, so
“`MUL r1, r0, #9`” is invalid syntax

`ADD r1, r0, r0, LSR #3`

; $r1 = r0 + r0 \gg 3 = r0 + r0/8$ (unsigned)

`ADD r1, r0, r0, ASR #3`

; $r1 = r0 + r0 \gg 3 = r0 + r0/8$ (signed)

Barrel Shifter

Logical Shift Left (**LSL**)



Logical Shift Right (**LSR**)



Rotate Right Extended (**RRX**)



Arithmetic Shift Right (**ASR**)



Rotate Right (**ROR**)



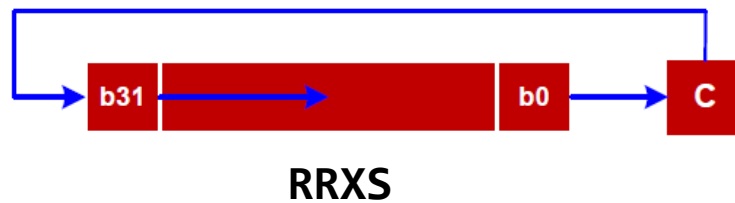
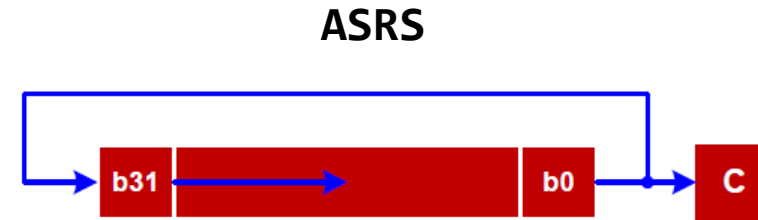
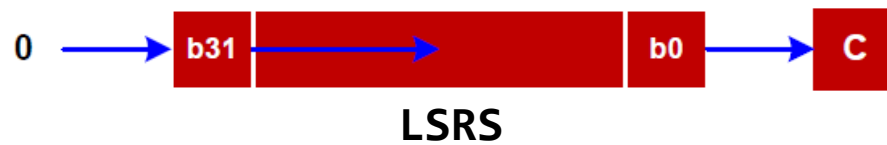
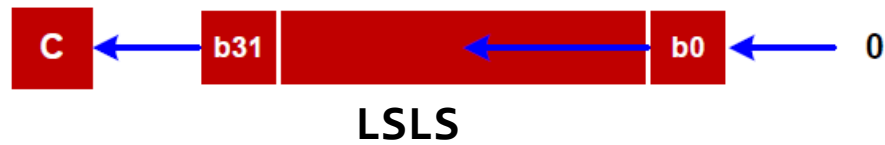
Why is there rotate right but no rotate left?

Rotate left can be replaced by a rotate right with a different rotate offset.

Updating APSR Flags

- If “S” is present, the instruction update flags. Otherwise, the flags are not updated.
- Let R be the final 32-bit result

N	Z	C	V
R<31>	IsZeroBit(R)	carry	unchanged



Barrel Shifter: Explanations

- ▶ LSL (logical shift left): **shifts left, fills zeros on the right**; C gets the last bit shifted out of bit 31. This is multiply by 2^n .
- ▶ LSR (logical shift right): **shifts right, fills zeros on the left**; C gets the last bit shifted out of bit 0. This is unsigned division by 2^n .
- ▶ ASR (arithmetic shift right): **shifts right, fills the sign bit on the left** to preserving the sign; C gets the last bit shifted out of bit 0. This is signed division by 2^n with sign extension
- ▶ ROR (rotate right): **rotates bits right with wraparound**; bits leaving bit 0 re-enter at bit 31, and C gets the bit that was rotated from bit 0 to bit 31. This is a pure rotation without data loss.
- ▶ RRX (rotate right extended): **rotates right by one through the carry flag**, treating C as a 33rd bit; new bit 31 comes from old C, and C receives old bit 0.

Examples (shifting by 4)

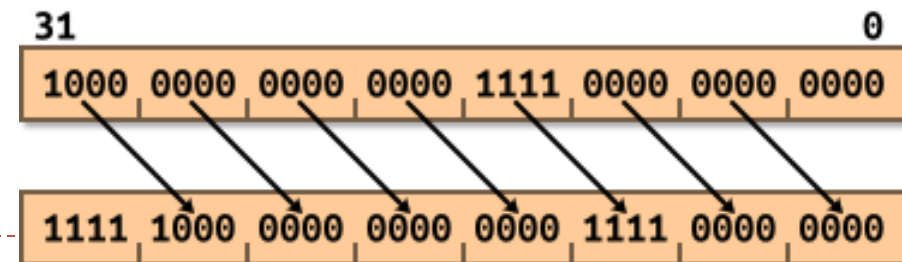
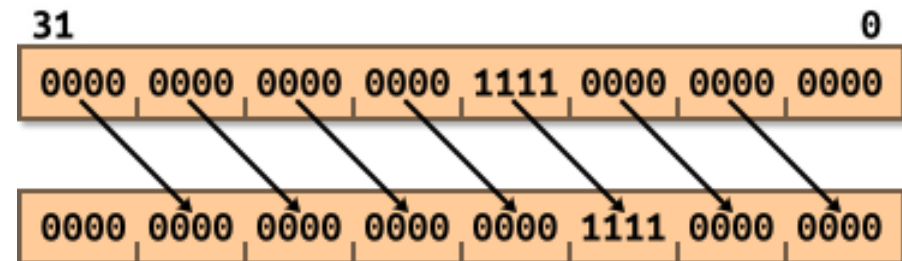
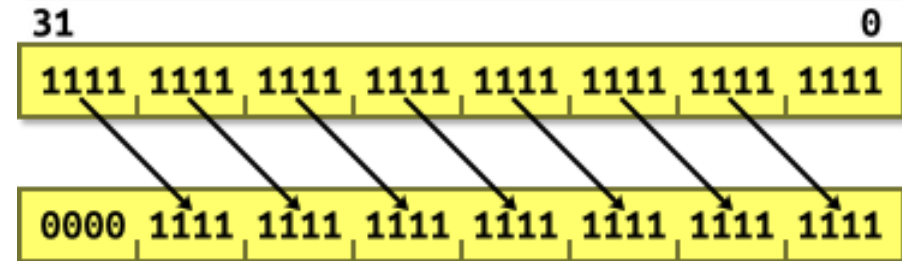
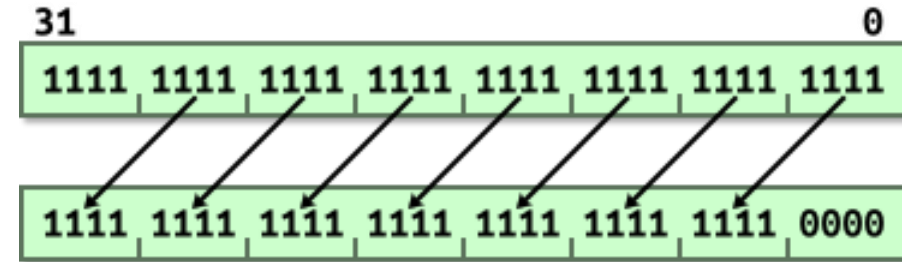
Logical Shift Left (**LSL**)



Logical Shift Right (**LSR**)

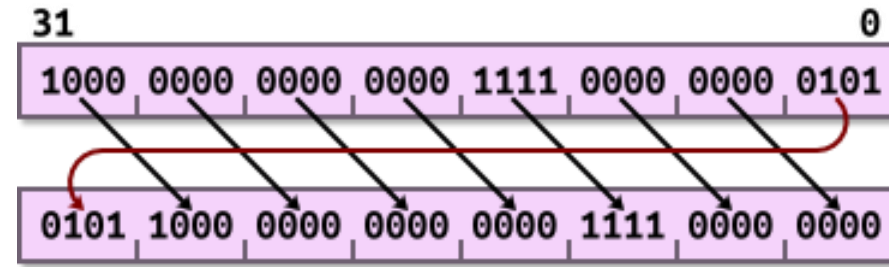


Arithmetic Shift Right (**ASR**)

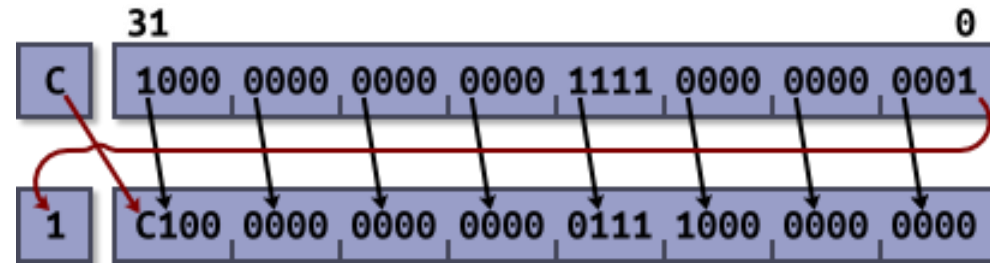
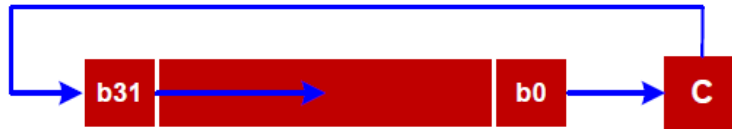


Examples (rotate)

Rotate Right (**ROR**) (rotate by 4)



Rotate Right Extended (**RRX**)
(can only rotate by 1)



Shift Operations

Logical Shift Left (LSL)



LSL {S} Rd, Rn, <shift>

moves all the bits of a register by n positions to the **left** and inserts n zeros in the right end

$$0 \leq n \leq 31$$

Example 1

```
; r2 = 0x0000_0001 (#1)
```

LSL r3, r2, #3

```
; r3 = 0x0000_0008 (#8)
```

```
; 8 = 23 * 1
```

Example 2

```
; r2 = 0x0000_0003 (#3)
```

LSL r3, r2, #2

```
; r3 = 0x0000_000C (#12)
```

```
; 12 = 22 * 3
```

Example 3

```
; r3 = 0xFFFF_0000 (#-65536)
```

LSLS r2, r3, #1

```
; r2 = 0xFFFE_0000 (#-131072)
```

```
; -131072 = 21 * -65536
```

C=1, N=1, Z=0, V=not updated

Note: If the suffix S is used, the carry flag is updated to the value of the last shifted bit.

Shift Operations

Logical Shift Right (LSR)



LSR{S} Rd, Rn, <shift>

moves all the bits of a register by n positions to the right and inserts n zeros in the left end

$$1 \leq n \leq 32$$

Example 1

```
; r2 = 0x0000_0010 (#16)
```

```
LSR r1, r2, #3
```

```
; r1 = 0x0000_0002 (#2)
```

```
; 2 = 16/23
```

Example 2

```
; r2 = 0x8000_0000 (# -2,147,483,648)
```

```
LSR r2, r2, #2
```

```
; r2 = 0x2000_0000 (# 536,870,912)
```

```
; 536,870,912 = -2,147,483,648/22
```

➡ with LSR sign bit is lost !

Example 3

```
; r2 = 0x0000_0001 (#1)
```

```
LSRS r3, r2, #1
```

```
; r3 = 0x0000_0000 (#0)
```

```
; 0 = 1/21
```

C=1, N=0, Z=1, V=not updated

Note: If the suffix S is used, the carry flag is updated to the value of the last shifted bit.

Shift Operations

Arithmetic Shift Right (ASR)



ASR{S} Rd, Rn, <shift>

moves all the bits of a register by n positions to the **right** and inserts n copies of the sign bit in the left end

$$1 \leq n \leq 32$$

Example 1

```
; r0 = 0xFFF8_0000 (-524288)
```

```
ASR r1, r0, #3
```

```
; r1 = 0xFFFF_0000 (-65536)
```

```
; -65536 = -524288/23
```

ASR is equivalent to signed integer division

Example 2

```
; r2 = 0x8000_0000 (-2,147,483,648)
```

```
ASR r2, r2, #2
```

```
; r2 = 0xE000_0000 (# -536,870,912)
```

```
; -536,870,912 = -2,147,483,648/22
```

Example 3

```
; r2 = 0xFFFF_F001 (#-4095)
```

```
ASRS r3, r2, #1
```

```
; r3 = 0xFFFF_F800 (#-2048)
```

```
; -2048 = -4096/21
```

C=1, N=1, Z=0, V=not updated

Note: If the suffix S is used, the carry flag is updated to the value of the last shifted bit.

Rotate Operations

Rotate Right (ROR)



ROR{S} Rd, Rn, <shift>

Circular shifts of all the bits of a register by n positions to the **right** as if the right end of the register is joined with its left end. The last shifted bit updates the carry bit

$$1 \leq n \leq 31$$

Example 1

```
; r2 = 0x0008_0000
```

```
ROR r2, r2, #10
```

```
; r2 = 0x0000_0200
```

Example 2: rotate **left** by 12 bits

```
; r0 = 0xF000_0000
```

```
ROR r2, r0, #20
```

```
; r2 = 0x0000_0F00
```

Rotate left by m bits is equivalent to rotate right
ROR by $32-m$ bits

Example 3

```
; r2 = 0xF0F0_F001
```

```
; r1 = 0x0000_000E
```

```
RORS r3, r2, r1
```

```
; r3 = 0xC007_C3C3
```

C=1, N=1, Z=0, V=not updated

Note: If the suffix S is used, the carry flag is updated to the value of the last shifted bit.

Rotate Operations

Rotate Right Extended (RRX)



RRX{S} Rd, Rn

This is a one-bit rotate instruction.

Example 1

```
; r2 = 0x0008_0003, c = 1
```

```
RRX r2, r2
```

```
; r2 = 0x8004_0001, c = 1
```

Example 2:

```
; r2 = 0xF000_0001, c = 0
```

```
RRX r1, r2
```

```
; r1 = 0x7800_0000, c = 0
```

Example 3

```
; r2 = 0xF0F0_F001, c = 0
```

```
RRXS r3, r2
```

```
; r3 = 0x7878_7800, c = 1
```

C=1, N=0, Z=0, V= not updated

Note: the carry flag is updated by b0 only if the suffix S is used, otherwise it keeps its original value

Barrel Shifter More Examples

- ▶ MOV r0, r0, LSL #1
 - ▶ $r0 = r0 * 2$
- ▶ MOV r1, r1, LSR #2
 - ▶ $r1 = r1 / 4$ (unsigned).
- ▶ MOV r2, r2, ASR #2
 - ▶ $r2 = r2 / 4$ (signed).
- ▶ MOV r3, r3, ROR #16
 - ▶ Swap the top and bottom halves of r3.
- ▶ ADD r4, r4, r4, LSL #4
 - ▶ $r4 = r4 * 17 (= r4 + r4 * 16)$
- ▶ RSB r5, r5, r5, LSL #5
 - ▶ $r5 = r5 * 31 (= r5 * 32 - r5)$
 - ▶ Reverse-subtract using barrel shifter on 2nd operand
- ▶ SUB r5, r5, r5, LSR #5
 - ▶ $r5 = r5 - (r5 / 32)$
- ▶ LDR r9, [r12, r8, LSL #2]
 - ▶ Load a 32-bit word into r9 from the memory address computed as $r12 + (r8 * 4)$

SUB vs. RSB

- ▶ SUB instruction: SUB Rd, Rn, Operand2 performs $Rd = Rn - \text{Operand2}$
- ▶ RSB instruction: RSB Rd, Rn, Operand2 performs $Rd = \text{Operand2} - Rn$
- ▶ There are equivalent:
 - ▶ SUB R5, R3, #10 @ R5 = R3 - 10
 - ▶ RSB R5, R3, #10 @ R5 = 10 - R3
- ▶ When to use RSB?
- ▶ Subtracting from constants, since constants can only appear as Operand2 in ARM instructions. For example:
 - ▶ RSB R2, R4, #1 means $R2 = 1 - R4$
 - ▶ This cannot be done with SUB without first loading the constant into a register
- ▶ Negation Operations by subtracting from zero:
 - ▶ RSB R0, R0, #0 effectively computes $R0 = 0 - R0 = -R0$
- ▶ Complex Operand2 Operations
 - ▶ RSB is valuable when you want to perform operations on Operand2 before subtraction, such as shifting :
 - ▶ RSB R1, R2, R3, LSL #1 computes $R1 = (R3 \ll 1) - R2$
 - ▶ This allows you to shift a value and then subtract from it in a single instruction

Integer Array Access with LSL

- ▶ To calculate the address of element `array[i]` of 32-bit integers, we calculate (base address of array) + $i*4$ for an array of words. For example:
 - ▶ `ADR r3,ARRAY` @ load base address of ARRAY into r3 (ARRAY contains 4-byte ints)
 - ▶ `MOV r2,#6` @ Suppose we want to access ARRAY[6]
 - ▶ `MOV r4,r2,LSL #2` @ logical shift i's value in r2 by 2 to multiply its value by 4
 - ▶ `ADD r5,r3,r4` @ finish calculation of the address of element array[i] in r5
 - ▶ `LDR r6,[r5]` @ load value of array[i] into r6 using the address in r5
- ▶ Alternatively, we can perform this same address calculation with a single ADD:
 - ▶ `ADD r5,r3,r2,LSL #2` @ calculate address of array[i] in r5 with single ADD
 - ▶ `LDR r6,[r5]` @ load value of array[i] into r4 using the address in r5
- ▶ Alternatively, ARM has some nice addressing modes to speedup array item access:
 - ▶ `LDR r6,[r3,r2,LSL #2]`

Example 1: ANDS

```
LDR  r0, =0xFFFFFFFF00
LDR  r1, =0x00000001
ANDS r2, r1, r0, LSL #1
```

Updates carry flag,
since ANDS does not update carry flag

N = 0, Z = 1, C = 1, V = not updated

AND{S}<c><q> {<Rd>,<Rn>, <Rm> {,<shift>}}

r0 = 0xFFFFFFFF00

r1 = 0x00000001

r0, LSL #1 = 0xFFFFFE00

r2 = r1 AND (r0 << 1) = 0x00000001 AND 0xFFFFFE00 = 0x00000000

ANDS sets flags:

Z = 1 (result r2 is zero)

N = 0 (bit 31 of result r2 is 0)

C is unaffected by ANDS, since logical operations don't affect overflow. It was set by previous shift "r0, LSL #1" to be C=1

V is unaffected by ANDS or shift (left unchanged from its previous value)



Example 2: ADDS

```
LDR  r0, =0xFFFFFFFF00
LDR  r1, =0x00000001
ADDS r2, r1, r0, LSL #1
```

Does NOT update carry flag,
since ADDS updates flags

N = 1, Z = 0, C = 0, V = 0

ADD{S}<c><q> {<Rd>,<Rn>, <Rm> {,<shift>}}

r0 = 0xFFFFFFFF00

r1 = 0x00000001

r0, LSL #1 = 0xFFFFFE00

r2 = r1 + (r0 << 1) = 0x00000001 + 0xFFFFFE00 = 0xFFFFFE01

ADDS sets flags:

Z = 0 (result r2 is non-zero)

N = 1 (bit 31 of result r2 is 1)

C = 0 (there is no carry out from bit 31 for unsigned addition, when adding 0x00000001 and 0xFFFFFE00)

V = 0 (there is no overflow for signed addition, when adding 0x00000001 and 0xFFFFFE00. Recall: adding a positive (1) to a negative (0xFFFFFE00) cannot cause overflow.)

Set a Bit in C

$$a \mid= (1 \ll k)$$

or

$$a = a \mid (1 \ll k)$$

Example: $k = 5$

a	a_7	a_6	a_5	a_4	a_3	a_2	a_1	a_0
$1 \ll k$	0	0	1	0	0	0	0	0
$a \mid (1 \ll k)$	a_7	a_6	1	a_4	a_3	a_2	a_1	a_0

The other bits should not be affected.

Set a Bit in Assembly

a |= (1 << 5)

Solution 1:

```
MOV r4, #1          ; r4 = 1
LSL r4, r4, #5       ; r4 = 1<<5
ORR r0, r0, r4      ; r0 = r0 | 1<<5
```

Solution 2:

```
MOV r4, #1          ; r4 = 1
ORR r0, r0, r4, LSL #5 ; r0 = r0 & not (1<<5)
```

Solution 3:

```
ORR r0, r0, # (1 << 5) ; r0 = r0 & not (1<<5)
```

Clear a Bit in C

$$a \ \&= \ \sim(1 \ll k)$$

Example: $k = 5$

a	a ₇	a ₆	a ₅	a ₄	a ₃	a ₂	a ₁	a ₀
$\sim(1 \ll k)$	1	1	0	1	1	1	1	1
$a \ \& \ \sim(1 \ll k)$	a ₇	a ₆	0	a ₄	a ₃	a ₂	a ₁	a ₀

The other bits should not be affected.

Clear a Bit in Assembly

a &= ~(1<<5)

Solution 1:

```
MOV r4, #1           ; r4 = 1
LSL r4, r4, #5        ; r4 = 1<<5
MVN r4, r4            ; r4 = not (1<<5)
AND r0, r0, r4        ; r0 = r0 & not (1<<5)
```

Solution 2:

```
MOV r4, #1           ; r4 = 1
MVN r4, r4, LSL #5    ; r4 = not (1<<5)
AND r0, r0, r4        ; r0 = r0 & not (1<<5)
```

Solution 3:

```
MOV r4, #1           ; r4 = 1
BIC r0, r0, r4, LSL #5 ; r0 = r0 & not (1<<5)
```

Solution 4:

```
BIC r0, r0, # (1 << 5) ; r0 = r0 & not (1<<5)
```



Toggle a Bit in C

Without knowing the initial value, a bit can be toggled by XORing it with a “1”

$$a \oplus= 1 \ll k$$

Example: $k = 5$

a	a_7	a_6	a_5	a_4	a_3	a_2	a_1	a_0
$1 \ll k$	0	0	1	0	0	0	0	0
$a \oplus (1 \ll k)$	a_7	a_6	NOT(a_5)	a_4	a_3	a_2	a_1	a_0

Truth table of
Exclusive OR

m	n	$m \oplus n$
0	0	0
0	1	1
1	0	1
1	1	0

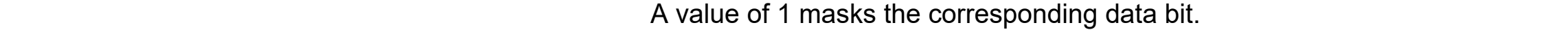
Toggle a Bit in Assembly

$a \oplus= 1 \ll 5$

Solution:

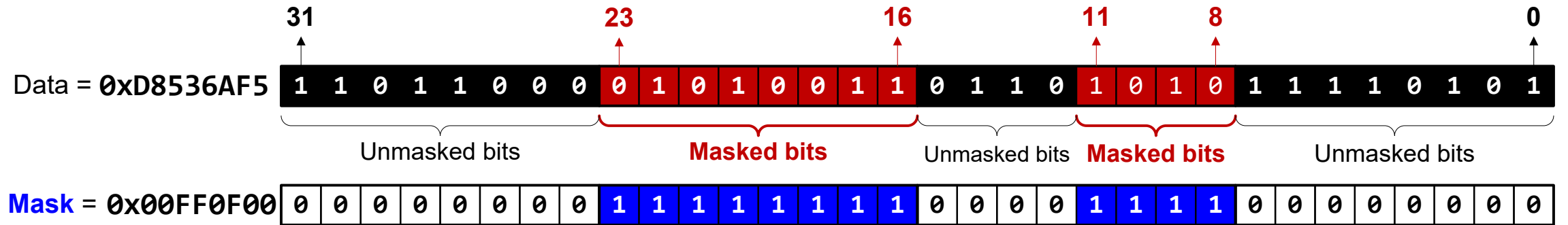
```
MOV r4, #1           ; r4 = 1
EOR r0, r0, r4, LSL #5 ; r0 = r0 ^ 1<<5
```

Here we can use MOVS and EORS instead of MOV and EOR, if the flags are used by later instructions.

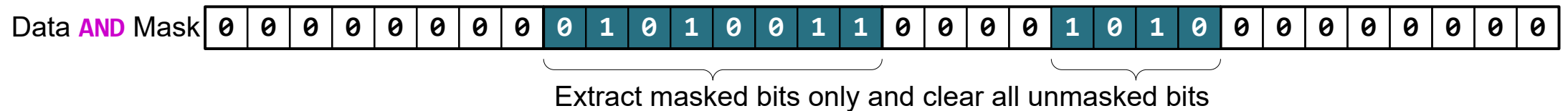


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Clear all unmasked bits

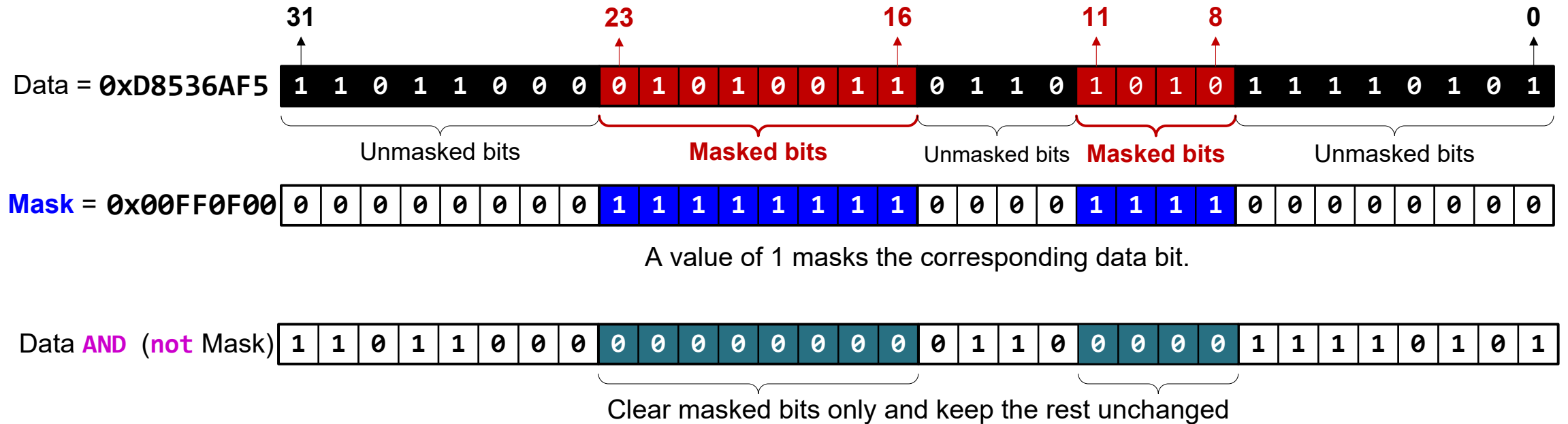


A value of 1 masks the corresponding data bit.



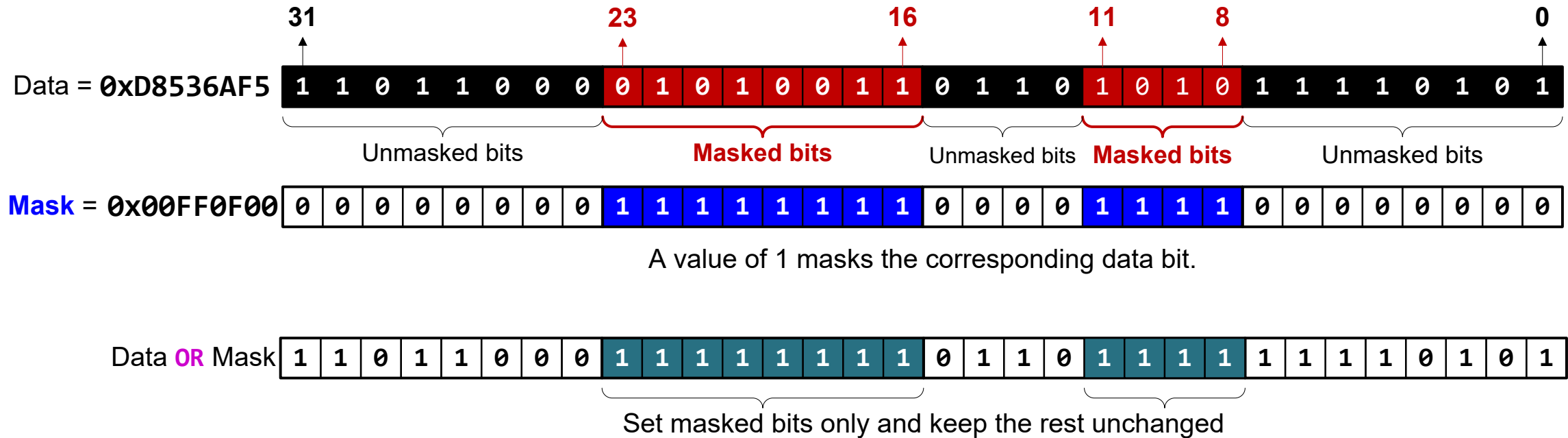
Data &= Mask;

Clear all masked bits

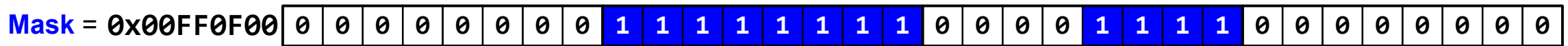


Data &= ~Mask;

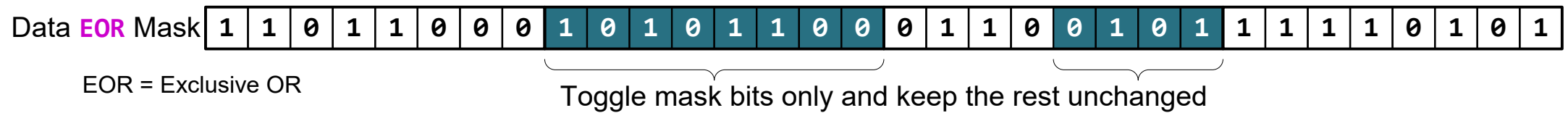
Set all masked bits



Data |= Mask;



A value of 1 masks the corresponding data bit.



Data ^= Mask;

Carry and Overflow Flags w/ Arithmetic Instructions

Carry flag $C = 1$ (Borrow flag = 0) upon an **unsigned** addition if the answer is wrong (true result $> 2^n - 1$)

Carry flag $C = 0$ (Borrow flag = 1) upon an **unsigned** subtraction if the answer is wrong (true result < 0)

Overflow flag $V = 1$ upon a **signed** addition or subtraction if the answer is wrong (true result $> 2^{n-1} - 1$ or true result $< -2^{n-1}$)

Overflow may occur when adding 2 operands with the same sign, or subtracting 2 operands with different signs; Overflow cannot occur when adding 2 operands with different signs or when subtracting 2 operands with the same sign.

	Unsigned Addition	Unsigned Subtraction	Signed Addition or Subtraction
Carry flag	true result $> 2^n - 1 \rightarrow$ Carry flag = 1 Borrow flag = 0 (Result incorrect)	true result $< 0 \rightarrow$ Carry flag = 0 Borrow flag = 1 (Result incorrect)	N/A
Overflow flag	N/A	N/A	true result $> 2^{n-1} - 1$ or true result $< -2^{n-1}$ $\rightarrow$ Overflow flag = 1 (Result incorrect)

Example

- ▶ For an 8-bit system, calculate $0x35 + 0x19$, setting C and V flags
- ▶ ANS:
 - ▶ Convert to binary and perform addition as in table
 - ▶ $C = 0$ since there is no carry-out from MSB b7
 - ▶ $V = 0$ since Op1, Op2 and result are all positive (sign bit = 0)
 - ▶ In decimal (not needed for exam): unsigned: $53 + 25 = 78$; signed: $53 + 25 = 78$

	0	1	1	0	0	0	1		Carry
	0	0	1	1	0	1	0	1	Op1: 0x35
	0	0	0	1	1	0	0	1	Op2: 0x19
C = 0									
V = 0	0	1	0	0	1	1	1	0	Result: 0x4E

Example

- ▶ For an 8-bit system, calculate $0x35 + 0x5B$, setting C and V flags
- ▶ ANS:
 - ▶ Convert to binary and perform addition as in table
 - ▶ $C = 0$ since there is no carry-out from MSB b7
 - ▶ $V = 0$ since Op1, Op2 are positive, result is negative
 - ▶ In decimal: unsigned: $53 + 91 = 144$; signed: $53 + 91 = -112$ (incorrect)

	1	1	1	1	1	1		Carry
	0	0	1	1	0	1	0	Op1: 0x35
	0	1	0	1	1	0	1	Op2: 0x5B
C = 0	1	0	0	1	0	0	0	Result: 0x90
V = 1								

Example

- ▶ For an 8-bit system, calculate $0x35 - 0x2D$, setting C and V flags
- ▶ ANS:
 - ▶ Convert to binary and perform addition as in table (another way is to perform subtraction in binary, but we do not cover it here)
 - ▶ $0x2D = 00101101$, its negation $TC(00101101) = 11010011 = 0xD3$
 - ▶ $C = 1$ since there is carry-out from MSB b7
 - ▶ $V = 0$ since Op1 is positive, Op2 is negative, result is negative
 - ▶ Overflow cannot occur when adding 2 operands with different signs
 - ▶ In decimal: unsigned: $53 + 211 = 8$ (incorrect); signed: $53 - 45 = 8$

$C = 1$ $V = 0$	1	1	1	0	1	1	1	Carry
	0	0	1	1	0	1	0	Op1: 0x35
	1	1	0	1	0	0	1	Op2: 0xD3
	0	0	0	0	1	0	0	Result: 0x08 (drop 1 in 0x108)

Example

- ▶ For an 8-bit system, calculate $-0x62 - 0x2D$, setting C and V flags
- ▶ ANS:
 - ▶ Convert to binary and perform addition as in table
 - ▶ $0x62 = 01100010$, its negation $TC(01100010) = 10011110 = 0x9E$
 - ▶ $0x2D = 00101101$, its negation $TC(00101101) = 11010011 = 0xD3$
 - ▶ $C = 1$ since there is carry-out from MSB b7
 - ▶ $V = 1$ since Op1, Op2 are both negative, result is positive
 - ▶ In decimal: unsigned: $158 + 211 = 113$ (incorrect); signed: $-98 - 45 = 113$ (incorrect)

$C = 1$ $V = 1$	0	0	1	1	1	1	0	Carry
	1	0	0	1	1	1	0	Op1: 0x9E
	1	1	0	1	0	0	1	Op2: 0xD3
	0	1	1	1	0	0	0	Result: 0x71 (drop 1 in 0x171)

Dec	Hex	Binary
53	35	0 1 1 0 0 0 1
+25	+19	0 0 1 1 0 1 0 1
78	4E	0 0 0 1 1 0 0 1
		0 1 0 0 1 1 1 0
		C = 0
		V = 0

Note no carry from bit 6 to bit 7 and no carry from bit 7 to C.

Dec	Hex	Binary
53	35	1 1 1 1 1 1 1
+91	+5B	0 0 1 1 0 1 0 1
144	90	0 1 0 1 1 0 1 1
		1 0 0 1 0 0 0 0
		C = 0
		V = 1

Note carry from bit 6 to bit 7 but no carry from bit 7 to C.

Thinking SIGNED we added two positive numbers and got a negative result. This can't be correct! Therefore, the OVERFLOW bit, V, is set to 1. Correct answer (144) is outside the range -128 to +127.

Dec	Hex	Binary
53	35	1 1 1 0 1 1 1
-45	+D3	0 0 1 1 0 1 0 1
8	108	1 1 0 1 0 0 1 1
		0 0 0 0 1 0 0 0
		C = 1
		V = 0

Ignore carry

Note carry from bit 6 to bit 7 and carry from bit 7 to C.

Thinking SIGNED we added a positive number to a negative number and got the correct positive answer. Therefore, the OVERFLOW bit, V, is cleared to 0. Correct answer (8) is inside the range -128 to +127.

Dec	Hex	Binary
-98	9E	0 0 1 1 1 1 0
-45	+D3	1 0 0 1 1 1 1 0
-143	171	1 1 0 1 0 0 1 1
		1 1 1 0 0 0 1
		C = 1
		V = 1

Ignore carry

Note no carry from bit 6 to bit 7 but there is a carry from bit 7 to C.

Thinking SIGNED we added two negative numbers and got a positive answer. This must be wrong! Therefore, the OVERFLOW bit, V, is set to 1. Correct answer (-143) is outside the range -128 to +127.

References

- ▶ Lesson 45b - Adders Carry and Overflow, LBEbooks
 - ▶ https://www.youtube.com/watch?v=9cXe_T99nL4
- ▶ Lecture 25. Arithmetic and Logical Instructions
 - ▶ <https://www.youtube.com/watch?v=H-vOP2yRUj4&list=PLRjhV4hUhlymmp5CCelFPyxbknsdcXCc8&index=25>
- ▶ Lecture 26. Updating NZCV bit flags
 - ▶ https://www.youtube.com/watch?v=SGjibMID2_A&list=PLRjhV4hUhlymmp5CCelFPyxbknsdcXCc8&index=26